IEEE Tunisia Section Entrepreneurship Local Group

TRC TechResolve Challenge 3.0

IEEE CN CONSULTANTS NETWORK TUNISIA SECTION

TechResolve Challenge 3.0

Phase 1 – Ideation & Problem Definition

1. Challenge Theme

AI for Smarter Campus Life

Universities are dynamic ecosystems where thousands of students, professors, and staff interact daily with infrastructure, services, and digital systems. Despite this complexity, many everyday challenges on campus remain inefficiently managed or lack innovative technological solutions.

The TechResolve Challenge 3.0 invites participants to design **AI-powered** solutions that enhance campus life by addressing **real challenges** within university environments.

This challenge aligns with the following SDGs:

- **SDG 13** – Climate Action
- **SDG 15** – Life on Land

Participants are expected to develop innovative concepts that combine technology, sustainability, and responsible innovation. The proposed solutions should demonstrate a **balance** between problem understanding, entrepreneurial thinking, and technical feasibility.

2. Challenge Areas

Projects should focus on solving practical problems related to campus environments, such as:

- Campus Safety and Emergency Management
- Environmental Sustainability and Energy Efficiency
- Smart Resource and Infrastructure Management
- Student Well-being and Support Systems
- Smart Mobility and Transportation within Campus
- Responsible Use of Digital Technologies

3. Solution Design

In this phase, teams must clearly define the problem they want to solve and propose a structured solution.

Teams should focus on:

- Identifying a relevant campus problem
- Understanding the target users
- Designing a value-driven solution
- Explaining the role of AI in the solution

At the end of this phase, teams should have:

- A clear problem statement
- A structured solution concept
- A defined AI approach

This phase evaluates innovation, relevance, and entrepreneurial thinking.

4. Ideation Submission

Each team must submit a short description of their proposed idea. The submission should include:

- Problem statement
- Target users
- Proposed solution concept
- Brief explanation of how AI will be used

This submission allows organizers and mentors to review the ideas and ensure they align with the challenge objectives.

Submission Deadline: March 15, 2026 – 23:59 H

This preparation will help teams move efficiently into the technical development phase during the hackathon.

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IEEE SMC Society – ISIMS IEEE Student Branch

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Phase 2 – Technical Development & Proof of Concept

1. Objective

During this phase, teams will develop a functional Proof of Concept **(PoC)** demonstrating the technical feasibility of their solution.

The goal is not to build a complete product, but to show how the core idea works in practice.

Possible implementations include:

- AI or Machine Learning models
- Data analysis and prediction systems
- Intelligent automation tools
- Computer vision or NLP applications
- Web or mobile interfaces connected to AI services

2. Technical Implementation

Teams are encouraged to integrate AI or intelligent digital technologies into their solutions to address the identified campus problem.

Participants are free to choose **the tools, programming languages, frameworks and platforms** that best support their solution.

Possible technical approaches may include, but are not limited to:

- Data analysis and intelligent decision support
- Predictive or analytical models
- Smart monitoring or automation systems
- Digital platforms supporting campus services
- Interactive systems that improve user experience or communication

The focus should be placed on **clarity, feasibility and functionality** of the prototype.

3. Deliverables

At the end of the challenge, each team must provide the following:

1. Solution Pitch Deck

A presentation describing:

- The identified problem
- Target users
- Proposed solution
- AI methodology used
- Potential impact

2. Technical Proof of Concept

A functional prototype demonstrating the core concept of the solution. This may include:

- A working AI model
- A prototype application
- A system architecture
- A simulation or demo

3. Final Pitch

Each team will present their solution to the jury, including:

- 5-minute presentation
- 3-minute Q&A session

Final submission deadline:
March 17, 2026 – 10:00 AM

After submission, teams will present their projects to the jury.

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Phase 2 – Technical Development & Proof of Concept

4. Total Score

Category	Score
Problem Relevance & Understanding	15
Innovation & Creativity	20
AI Integration & Technical Quality	25
Proof of Concept Functionality	20
Impact & Scalability	10
Pitch & Presentation	10